

One-sided TD-residual clipping in the appetite-regulation MDP: why the *sign* of the clip determines the *direction* of the intake bias

Abstract

Ablation 1 reports that a negative one-sided clip on the TD residual ($c < 0$, flooring worse-than-expected outcomes) produces sustained *overconsumption*, while a positive clip ($c > 0$, capping better-than-expected outcomes) produces sustained *underconsumption*. This note derives that result rather than describing it. Three facts do the work. First, at the magnitudes used the clip is not a damper but a *half-wave rectifier*: it renders the advantage estimate sign-definite (Lemma 3). Second, because the reported configuration has a single menu slot, the policy is a Bernoulli field and the policy gradient along the eat/no-eat margin is *exactly* the advantage gap (Lemma 6) — no scalar-intensity approximation is required. Third, the clipped advantage enters the value target *without* the mean-centring applied to the actor’s copy, so the critic residual is sign-definite at every visited state and the induced value operator has no interior fixed point (Theorem 8). We then reduce the behavioural question to the sign of two measurable quantities, ΔU and ΔD — the effect of one meal on the discounted upward and downward surprise — and measure them by paired forking under common random numbers. The measurement is asymmetric, and the asymmetry is the result: $\Delta D > 0$ at *every* intake level, so the $c > 0$ undereating conclusion follows globally from the actor channel alone; $\Delta U > 0$ only near the regulating policy, so the $c < 0$ overeating *runaway* cannot be actor-driven and must be attributed to the critic channel. This predicts a decisive two-line ablation that separates the channels.

1 Setup and notation

We work throughout in the reported configuration: one menu slot ($K = 1$), so the action set is binary,

$$a_t \in \{0, 1\}, \quad a_t = 0 \text{ eat nothing}, \quad a_t = 1 \text{ eat the offered item at the fixed fraction } \nu = 0.6. \quad (1)$$

The policy is therefore a Bernoulli field parameterised by a logit,

$$\pi_\phi(1 \mid o) = p_\phi(o) = \sigma(z_\phi(o)) \in (0, 1), \quad (2)$$

with $p_\phi(o)$ bounded away from 0 and 1 by the entropy bonus. Write $x_t \in \mathbb{R}^3$ for the three regulated readouts (glucose, peptides, fatty acids). With $w_i = 1$ the readout coincides with the leaky-integrator state, and since that integrator is *linear*, a meal eaten at time t adds an exactly known perturbation to the entire future path:

$$x_{t+k}^{(a_t=1)} = x_{t+k}^{(a_t=0)} + \nu \varphi_i^f(k), \quad \varphi_i^f(k) = \sum_{j \leq k} (1 - d_i)^{k-j} p_i^f(j), \quad (3)$$

where p_i^f is the per-step absorption profile of food f on nutrient i , d_i the phase-dependent decay rate, and φ_i^f the resulting impulse response. Equation (3) holds exactly, for all k , holding future actions fixed; it is the reason a single ingestion event can be reasoned about in closed form.

The reward is the piecewise band function of the methodology (there Equation 11): $r(x) = \sum_i \rho_i(x_i)$ with $\rho_i = w_i \beta_i$ inside $[l_i, h_i]$, $-2w_i(x_i - h_i)^2$ above, and $-2w_i \sigma_i(l_i - x_i)^2$ below ($\sigma_i > 1$ during sleep). The TD residual, the one-sided clip and the GAE recursion are as in the methodology:

$$\delta_t = r_t + \gamma V(o_{t+1}) - V(o_t), \quad \delta'_t = \begin{cases} \min(\delta_t, c), & c > 0, \\ \max(\delta_t, c), & c < 0, \\ \delta_t, & c = 0 \text{ or None,} \end{cases} \quad \hat{A}_t = \sum_{l \geq 0} (\gamma \lambda)^l \delta'_{t+l}. \quad (4)$$

Throughout, $\gamma = 0.99$, $\lambda = 0.95$, so $\gamma \lambda = 0.9405$ and the GAE horizon is

$$H = \frac{1}{1 - \gamma \lambda} = 16.8 \text{ steps} = 168 \text{ simulated minutes}. \quad (5)$$

Two functionals of the residual stream carry the whole argument. Define the discounted *upward* and *downward* surprise

$$U_t = \sum_{l \geq 0} (\gamma \lambda)^l [\delta_{t+l}]_+, \quad D_t = \sum_{l \geq 0} (\gamma \lambda)^l [\delta_{t+l}]_-, \quad (6)$$

with $[u]_+ = \max(u, 0)$ and $[u]_- = \max(-u, 0)$, so $U_t, D_t \geq 0$ and the unclipped advantage is $\hat{A}_t = U_t - D_t$. We further write $T_t = U_t + D_t$ for the *total surprise*, the discounted ℓ_1 total variation of the value process.

Remark 1 (Two channels, only one of which is normalised). In `agents/pppo.py`, `_compute_gae` builds the value target as

```
returns = advs + values[:-1]
```

on the *un-normalised* clipped advantage, after which the caller mean-centres and rescales only the actor's copy:

```
adv_s = (adv_s - adv_s.mean()) / (adv_s.std() + 1e-8)
```

The critic therefore consumes the raw biased advantage and the actor a centred version. These two channels behave differently and are treated separately in Sections 5 and 4. Conflating them is the main way to get this analysis wrong.

2 Assumptions, and their measured status

Assumption 1 (Unidirectional actuation). $\varphi_i^f \geq 0$ in aggregate: for every (food, nutrient) pair $\sum_k p_i^f(k) \geq 0$, with strict positivity in at least one nutrient per food. The agent possesses no action that removes nutrient; downward motion of x_t is exclusively the passive decay $(1 - d_i)$.

Verified. Over the 19-item pool, $\sum_k p_{\text{glucose}}^f(k) > 0$ for all 19 foods (range +0.136 to +0.335 on the normalised scale); for peptides and fatty acids the sum is ≥ 0 for all foods and exactly 0 for the four items with no measurable content of that macronutrient. No violations. Note that assumption 1 concerns the *aggregate*: the glucose profile itself is biphasic, with 66.5% of its per-step increments negative, which is what makes a single meal generate both upward and downward value movement.

Assumption 2 (Effective rectification). $|c| \ll \text{med } |\delta|$ over the operating regime.

Verified on the recorded residual logs (3.24×10^5 residuals per condition, `enable_td_logging`): at $c = -10^{-4}$, $\mathbb{P}(|\delta| < |c|) = 7 \times 10^{-4}$ against $\text{med } |\delta| = 3.07$; at $c = +10^{-4}$, $\mathbb{P}(|\delta| < |c|) = 0$ against $\text{med } |\delta| = 0.46$. The fraction of residuals actually modified by the clip is 97.7% and 98.0% respectively.

Table 1: Band geometry and actuator coarseness on the normalised readout scale. κ_i is Equation (7), median over the 19 foods, with the range across foods in brackets. Peak lag is the median $\arg \max_k \varphi_i^f(k)$. Computed at ten simulated minutes per step (Remark 2).

nutrient	l_i	target	h_i	$h_i - l_i$	κ_i [range]	peak lag
glucose	-0.0010	0.2751	0.5512	0.5522	0.48 [0.22, 1.03]	2 steps (20 min)
peptides	0.0838	0.2235	0.3631	0.2793	0.89 [0.15, 1.87]	4 steps (40 min)
fatty acids	0.2500	0.3571	0.4643	0.2143	1.25 [0.00, 2.58]	1 step (10 min)

2.1 Actuator coarseness

A third structural fact, measured rather than assumed, explains why a meal is intrinsically a high-total-variation event. Define the *coarseness ratio*

$$\kappa_i^f = \frac{\nu \max_k \varphi_i^f(k)}{h_i - l_i}, \quad (7)$$

the peak excursion caused by one $\nu = 0.6$ meal, in units of the target band width. Because the discrete action space offers no portion control, κ cannot be titrated: it is a fixed property of each (food, nutrient) pair.

Table 1 gives the values. The median κ for fatty acids exceeds unity: *the typical single meal, taken at the bottom of the fatty acid band, overshoots the top of it*. Since the action is quantised at $\nu = 0.6$ with no smaller option, there is in general no meal that lands the state inside the band and keeps it there. Every ingestion event therefore produces a rise followed by an overshoot and a decay — upward value movement followed by downward value movement — with the peak arriving after 1–4 steps, far inside the horizon $H = 16.8$ of Equation (5). This is the structural reason both U and D respond to eating, and it is what Section 6 measures directly.

Remark 2 (A discrepancy between the code and the methodology). The methodology states ten simulated minutes per environment step, which is the value consistent with its other claims: 144 steps per cycle is then exactly 24 h (96 awake = 16 h, 48 asleep = 8 h) and a 432-step episode is 72 h. The constant currently in `envs/env.py` is `MINUTES_PER_STEP = 5`, which would make a cycle 12 h and an episode 36 h. Table 1 is computed at ten minutes per step; at five the picture is qualitatively identical and marginally weaker ($\kappa = 0.48/0.83/1.12$, peak lags 4/8/2 steps). All conclusions below are robust to the choice (Section 6), but the constant should be reconciled before publication.

3 The clip is a rectifier, not a damper

Lemma 3 (Rectification). *Under Assumption 2, uniformly in t ,*

$$\left| \hat{A}_t - U_t \right| \leq \frac{|c|}{1 - \gamma\lambda} \quad (c < 0), \quad \left| \hat{A}_t + D_t \right| \leq \frac{|c|}{1 - \gamma\lambda} \quad (c > 0). \quad (8)$$

In particular the clipped advantage is sign-definite up to $O(|c|/(1 - \gamma\lambda))$, which at the reported magnitudes is $\approx 10^{-3}$:

$$\hat{A}_t \approx U_t \geq 0 \quad (c < 0), \quad \hat{A}_t \approx -D_t \leq 0 \quad (c > 0). \quad (9)$$

Proof. Take $c < 0$. If $\delta \geq 0$ then $\max(\delta, c) = \delta = [\delta]_+$ and the pointwise error vanishes. If $\delta < 0$ then $\max(\delta, c) \in [c, 0)$ while $[\delta]_+ = 0$, so the pointwise error is at most $|c|$. Hence $|\delta'_t - [\delta]_+| \leq |c|$ for all t , and summing the geometric weights in Equation (4) gives $|\hat{A}_t - U_t| \leq |c| \sum_{l \geq 0} (\gamma\lambda)^l = |c|/(1 - \gamma\lambda)$. The case $c > 0$ is the mirror image, comparing $\min(\delta, c)$ with $-[\delta]_-$. $\square$

Lemma 3 is confirmed directly by the recorded logs. The clipped residual stream spans $[-10^{-4}, +30.28]$ at $c = -10^{-4}$ and $[-5.33, +10^{-4}]$ at $c = +10^{-4}$: in each case the retained side is unbounded and the suppressed side is pinned at c . The estimator is not a mildly robustified advantage, but a one-sided functional.

Remark 4 (What has been substituted for what). For $\lambda \rightarrow 1$, $\gamma \rightarrow 1$ the GAE sum telescopes, $\hat{A}_t = G_t - V(o_t) = U_t - D_t$: the net change of the value process, i.e. upward movement minus downward movement. Lemma 3 says the clip deletes one of the two terms. A negative clip optimises upward movement alone — reward seeking with no punishment avoidance — and a positive clip optimises minus downward movement alone — punishment avoidance with no reward seeking. Everything that follows is a quantitative version of this sentence.

Lemma 5 (Decomposition). $U_t = \frac{1}{2}(T_t + \hat{A}_t^{\text{raw}})$ and $D_t = \frac{1}{2}(T_t - \hat{A}_t^{\text{raw}})$, where $\hat{A}_t^{\text{raw}} = U_t - D_t$ is the unclipped advantage. Consequently, for any two actions with gaps $\Delta U, \Delta D, \Delta T, \Delta A$,

$$[\Delta U > 0 \wedge \Delta D > 0] \iff \Delta T > |\Delta A|. \quad (10)$$

Equation (10) is the useful form: it says the clipped estimators point in a fixed direction, irrespective of the true advantage, exactly when the meal is *more surprising than it is on-net beneficial or harmful*.

4 Actor channel: the gradient along the eat/no-eat margin

Because the action set is binary, no scalar reduction is needed — the relevant gradient is available in closed form.

Lemma 6 (Bernoulli logit gradient). For the policy (2), $\nabla_z \log \pi(a | o) = a - p(o)$, hence for any advantage estimator $\hat{A}$,

$$\frac{\partial}{\partial z(o)} \mathbb{E}[\log \pi(a | o) \hat{A}(o, a)] = p(o)(1 - p(o)) [\hat{A}(o, 1) - \hat{A}(o, 0)]. \quad (11)$$

Combining with Lemma 3, and writing $\Delta U(o) = U(o, 1) - U(o, 0)$, $\Delta D(o) = D(o, 1) - D(o, 0)$,

$$\frac{\partial J}{\partial z(o)} \propto \begin{cases} +\Delta U(o), & c < 0, \\ -\Delta D(o), & c > 0, \end{cases} \quad \text{with } p(1 - p) > 0. \quad (12)$$

Equation (12) is an identity, not an approximation: the sign of the gradient on the probability of eating is the sign of ΔU (for $c < 0$) or of $-\Delta D$ (for $c > 0$). This is the step that the multi-slot ($K > 1$) case can only approximate, and it is available here precisely because the reported configuration has $K = 1$.

Lemma 7 (Mean-centring cannot repair a sign-definite advantage). Let $\tilde{A} = (\hat{A} - \bar{A})/(\hat{s} + \epsilon)$ with $\bar{A}, \hat{s}$ scalars computed from the batch. Since $\mathbb{E}_{a \sim \pi}[\nabla \log \pi(a | o)] = 0$, subtracting any quantity independent of (o, a) leaves the expected gradient unchanged, and dividing by $\hat{s} > 0$ rescales it by a positive constant. Hence normalisation alters neither the sign in Equation (12) nor the direction of the induced flow.

Lemma 7 is worth stating explicitly because the natural objection to Lemma 3 — “a uniformly non-negative advantage is re-centred anyway” — is false. Centring removes the offset; the bias lives in the *shape*, and Equation (12) shows the shape is what the gradient reads.

5 Critic channel: sign-definite value drift

Theorem 8 (No interior fixed point of the value target). *The regression target in `_compute_gae` is $\hat{R}_t = \hat{A}_t + V(o_t)$, formed on the un-normalised clipped advantage (Remark 1). By Lemma 3,*

$$\hat{R}_t - V(o_t) = \begin{cases} U_t \geq 0 & (c < 0), \\ -D_t \leq 0 & (c > 0), \end{cases} \quad \text{for every visited state.} \quad (13)$$

Since $\nabla_{\theta} L^V = -2\mathbb{E}[(\hat{R}_t - V_{\theta}(o_t))\nabla_{\theta} V_{\theta}(o_t)]$ and the residual is sign-definite across the whole batch, every gradient step moves V in one direction at every visited state. Define $\mathcal{T}_c V(o) = V(o) + \mathbb{E}[U_t \mid o_t = o]$ for $c < 0$. Then $\mathcal{T}_c V \geq V$ pointwise, with equality only on $\{o : \mathbb{P}(\exists l < H : \delta_{t+l} > 0 \mid o) = 0\}$, a null set under a policy with full support. Hence $\mathcal{T}_c$ admits no interior fixed point and iteration diverges: $V \rightarrow +\infty$ for $c < 0$ and $V \rightarrow -\infty$ for $c > 0$.

Corollary 9 (Predictable sign of critic miscalibration). *Under correct PPO, $\mathbb{E}[\delta_t] = 0$ by definition of V^{π} . Under a one-sided clip, $c < 0$ inflates V , so realised residuals become systematically negative, and $c > 0$ deflates V , so they become systematically positive.*

Measured. $\mathbb{E}[\delta] = -6.93$ with 97.7% of residuals negative at $c = -10^{-4}$; $\mathbb{E}[\delta] = +0.60$ with 98.0% positive at $c = +10^{-4}$. A mean residual of ∓ 7 where the correct value is 0 is an artefact of the clip alone, of exactly the predicted sign.

Corollary 10 (Measure degeneracy and instability). *The surviving residual tail carries $\approx 2\%$ of steps, so the gradient estimator loads on a rare high-magnitude tail and its variance grows. Median $|\delta|$ rises from 0.099 to 18.9 across 500 retraining episodes at $c = -10^{-4}$. The prediction is bimodality across seeds rather than uniform overeating, which is what the per-seed inference table shows (seed 7: +170 return, 11 items; seed 27: -6893 return, 191 items), and which the across-seed standard deviations confirm: 89.2 items at $c = -10^{-4}$ against 4.8 at $c = +10^{-5}$ and 0.8 with no clip.*

6 Numerical determination of ΔU and ΔD

By Lemma 6 the behavioural question reduces entirely to the signs of $\mathbb{E}[\Delta U]$ and $\mathbb{E}[\Delta D]$ as functions of the intake level. We measure them by *paired forking under common random numbers*. Fix a state-independent Bernoulli policy with eating probability θ while awake. For each of 1500 awake fork states o : snapshot the full environment state (in-progress meals, phase, menu); run one branch with $a = 1$ and one with $a = 0$; give both branches the *same* pre-drawn menu sequence and the *same* pre-drawn action-randomisation stream, so the branches differ only in the forked action. On each branch accumulate U and D over the GAE window from Equation (6), and additionally a value-function-free Monte-Carlo advantage gap

$$\Delta A_{\text{mc}} = \sum_k \gamma^k (r_k^{(1)} - r_k^{(0)}), \quad (14)$$

which under common random numbers estimates $Q(o, 1) - Q(o, 0)$ with no dependence on any value estimate. V^{π} itself is fitted by Monte-Carlo regression (random Fourier features and ridge) on $[x_t, \text{awake}, \text{phase}, \text{phase harmonics}, \text{remaining episode fraction}, \text{menu one-hot}]$; held-out R^2 is reported per condition.

We use a Bernoulli policy class, rather than a trained checkpoint, because this is the correct object for the question: ΔU and ΔD are properties of the *environment* evaluated at a policy, and the claim to be tested is that eating raises the total variation of the value process. The class also

Table 2: Paired-fork estimates. 1500 forks per row; all standard errors ≤ 0.18 . “regulated” is the fraction of fork states at or above the band (signed band error > -0.05). ΔA_{mc} is Equation (14) and is value-function-free. The row nearest θ^* is marked. Computed at five simulated minutes per step; see Remark 2 and the robustness note below.

θ	meals /ep	regu- lated	ΔU		ΔD		ΔA_{mc}	R^2
			mean	frac >0	mean	frac >0		
0.008	2	0.13	+13.95	0.989	+5.53	0.875	+8.58	0.50
0.020	6	0.33	+5.38	0.955	+2.16	0.693	+3.15	0.65
0.030*	9	0.49	+2.74	0.884	+1.96	0.655	+0.55	0.70
0.043	12	0.66	+1.20	0.737	+2.61	0.757	−1.64	0.66
0.100	29	0.90	−0.71	0.336	+4.58	0.957	−5.46	0.36
0.200	58	0.96	−1.30	0.365	+6.98	0.909	−8.36	0.48

brackets the empirical operating point: ΔA_{mc} changes sign at $\theta^* \approx 0.032$, or 9–10 meals per episode, against 12.5 items for the trained no-clip agent.

Table 2 reports the result, and it is asymmetric in a way that matters.

Robustness. The ΔU sign change was re-estimated with $3\times$ the regression data and $2\times$ the feature count, giving $-0.71 \rightarrow -0.71$ at $\theta = 0.100$: it is not a fitting artefact. Repeating the experiment at ten simulated minutes per step (Remark 2) gives $\Delta U = +3.43$ (0.893), $\Delta D = +2.31$ (0.693) at $\theta = 0.030$ and $\Delta U = -0.14$ (0.471), $\Delta D = +5.08$ (0.931) at $\theta = 0.100$ — the same qualitative pattern, with the ΔU zero-crossing displaced from $\theta \approx 0.06$ to $\theta \approx 0.10$. The regression fit is weakest in the high- θ rows ($R^2 = 0.36$ – 0.48), which is where the value process is largest in magnitude; those rows should be read as establishing a sign, not a magnitude.

6.1 Consequence for $c > 0$: undereating, from the actor channel alone

Proposition 11 (Monotone descent of intake). $\mathbb{E}[\Delta D] > 0$ at every θ examined, from $\theta = 0.008$ to $\theta = 0.200$, with standard errors ≤ 0.18 and $\mathbb{P}(\Delta D > 0) \geq 0.655$ throughout. By Equation (12) the gradient flow on the eating logit therefore satisfies

$$\dot{\theta} \propto -\mathbb{E}[\Delta D] < 0 \quad \text{for all } \theta \in [0.008, 0.200], \quad (15)$$

so the flow is monotone decreasing with no interior fixed point over the whole examined range. Intake collapses.

The $c > 0$ direction is thus fully explained by the actor channel, and the explanation is unconditional. Two refinements match the data quantitatively. First, ΔD is smallest near θ^* and grows in both directions away from it, so the suppression accelerates once the policy has been pushed off the regulating point. Second, and more usefully, ΔD is strongly regime-dependent: at $\theta = 0.043$ it is $+4.02$ ($\mathbb{P} = 0.912$) in regulated states but -0.14 ($\mathbb{P} = 0.457$) in depleted ones, a collapse of roughly $6\times$. The descent therefore *stalls* once the policy has driven the state distribution into depletion, predicting a non-zero intake floor rather than $\theta \rightarrow 0$. The observed floor is 4–9 items per episode ($\theta \approx 0.015$ – 0.031), consistent with this.

Remark 12. This corrects a natural but wrong first guess, namely that $c > 0$ should drive eating to zero. It does not, and the reason is visible in the measurement: the punishment-avoidance signal has no grip on a policy that is already starving, because a depleted trajectory is smooth and predictable and therefore surprise-free.

6.2 Consequence for $c < 0$: the actor channel only destabilises

Proposition 13 (Local instability, but no actor-driven runaway). $\mathbb{E}[\Delta U] > 0$ for $\theta \lesssim 0.06$ (five minutes per step) or $\theta \lesssim 0.10$ (ten minutes per step) and negative above it. Hence by Equation (12) the regulating policy θ^* is unstable under $c < 0$ — at θ^* we measure $\Delta U = +2.74$ with $\mathbb{P}(\Delta U > 0) = 0.884$, and among the fork states where eating is genuinely harmful ($\Delta A_{\text{mc}} < 0$) the negative clip still pushes towards eating in 84.0% of cases — but the flow possesses a stable interior fixed point at

$$\theta_{\text{fix}} \approx 0.06\text{--}0.10 \quad (\approx 18\text{--}29 \text{ meals per episode}). \quad (16)$$

Equation (16) is 1.5–2.4 $\times$ the no-clip intake, which is overeating, but the observed $c < 0$ behaviour is 114–139 items per episode ($\theta \approx 0.40\text{--}0.48$), some 5–7 $\times$ beyond θ_{fix} . The actor channel therefore accounts for the *direction* of the $c < 0$ effect and for the *loss of stability* of the regulating policy, but not for the magnitude of the runaway. The residual must be supplied by the critic channel: by Theorem 8 the value function is being driven upward without bound, and Corollary 9’s measured $\mathbb{E}[\delta] = -6.93$ confirms it. An inflated V keeps the realised residual stream overwhelmingly negative, so the 2% surviving positive tail remains attached to ingestion events and continues to reinforce them long past the point at which a calibrated critic would have arrested the flow.

Corollary 14 (The two directions fail by different mechanisms). $c > 0$ is a single-channel, monotone, actor-driven failure with a floor; $c < 0$ is a two-channel failure in which the actor destabilises the regulating policy and the diverging critic sustains the departure. This predicts the severity and stability asymmetry observed: $c < 0$ gives return -2631 with an across-seed dispersion of 89.2 items, while $c > 0$ gives return $+106$ with a dispersion of 4.8 items.

7 Independence from the reward asymmetry

The derivation used Assumptions 1 and 2, the coarseness measurement of Section 2.1, and Lemmas 3–7. It nowhere used the fact that the band reward has a bounded upside β_i and an unbounded quadratic downside, nor the sleep penalty $\sigma_i > 1$. This is worth emphasising because $\sigma_i \in \{1.5, 2.0\}$ makes depletion *strictly worse* than overshoot, which if anything biases the agent towards eating more — the opposite of the measured $c > 0$ result. The reward asymmetry cannot therefore be the mechanism. The mechanism is Assumption 1: the actuator is one-sided. Eating is the only way for the state to move up and waiting is the only way for it to move down, so “which side of the surprise is retained” maps one-to-one onto “which action is reinforced”.

8 Consistency with the magnitude sweep

Lemma 3 makes a sharp prediction that the existing sweep tests without further computation: the effect should be a *step function of sign(c) that saturates in $|c|$* , since once $|c| \ll \text{med } |\delta|$ the operator is already a pure rectifier and shrinking c further changes nothing; and it should *vanish* once $|c| \gtrsim \text{med } |\delta|$, where the clip stops binding.

Table 3 bears this out on both counts. On the negative side there is a flat plateau of 105–139 items across $|c| \in [10^{-6}, 10^{-4}]$ — four orders of magnitude with no graded dependence, as the rectification limit requires — a sharp threshold between -5×10^{-4} and -10^{-3} , and full recovery to no-clip behaviour for $|c| \geq 10^{-3}$, which is the $\text{med } |\delta|$ scale. The positive side recovers monotonically from 4.3 to 12.7 items over the same range. The no-clip condition sits between the two branches.

Table 3: Retraining sweep, 10 seeds per condition, mean over the final 50 of 500 retraining episodes. Selected rows from the 37 available conditions.

c	items/ep	return	c	items/ep	return
-5×10^{-3}	10.0	+165	$+1 \times 10^{-6}$	7.2	+100
-1×10^{-3}	11.4	+172	$+5 \times 10^{-6}$	4.3	+77
-5×10^{-4}	33.9	-543	$+1 \times 10^{-5}$	7.8	+106
-1×10^{-4}	114.3	-2631	$+1 \times 10^{-4}$	9.3	+116
-1×10^{-5}	137.3	-3346	$+1 \times 10^{-3}$	8.6	+116
-1×10^{-6}	137.0	-3108	$+5 \times 10^{-2}$	12.7	+154
$c = 0$ (no clip): 12.5 items/ep, return +159					

An exception worth reporting. At $c = +2 \times 10^{-4}$ and $+3 \times 10^{-4}$ the mean intake is 28.5 and 31.8 items with across-seed standard deviations of 60 and 79: a minority of seeds crossed into the overeating mode. A candidate explanation follows from Corollary 9: a positive clip deflates V , so residuals become overwhelmingly positive, and at intermediate $|c|$ nearly all of them exceed $+c$, leaving $\hat{A} \approx -D \approx 0$ and starving the gradient. This should be checked rather than assumed.

9 Predictions

P1 (decisive; two lines of code). Clip only the actor’s advantage and leave the critic’s target unclipped — that is, form `returns` in `_compute_gae` from the raw δ while still clipping `adv`s. By Propositions 11 and 13 this should collapse the $c < 0$ intake from ≈ 139 items to θ_{fix} ’s 18–29, while leaving the $c > 0$ undereating essentially unchanged. This isolates the two channels and converts Corollary 14 from an inference into a measurement.

P2. Log the mean value estimate during retraining: by Theorem 8 it must rise monotonically for $c < 0$ and fall for $c > 0$.

P3. Introduce a bidirectional actuator, i.e. an action with negative nutrient input, violating Assumption 1. The $c < 0$ bias should partly redirect onto that action and the intake effect should weaken. This is the direct test of the mechanism identified in Section 7.

P4. Make ingestion predictable (fixed meal schedule, no menu resampling). Both U and D shrink, and both clip effects should shrink towards zero.

P5. A *symmetric* clamp $|\delta| \leq c$ should be comparatively benign at the same magnitude, since it preserves $\text{sign}(\hat{A})$ and hence the sign in Equation (12).

P6. Add portion control, reducing κ in Equation (7) below unity for all nutrients. The overshoot component of D should fall and the $c > 0$ effect weaken.

10 Limitations

1. Table 2 uses a state-independent Bernoulli policy class, not the trained network. This is the appropriate object for testing a claim about the environment, and θ^* brackets the trained

agent’s intake, but ΔU and ΔD at the network’s actual state distribution are not measured.

2. V^π is fitted, with held-out R^2 of 0.65–0.70 near θ^* but only 0.36–0.48 at high θ . The high- θ rows establish signs, not magnitudes. The ΔA_{mc} column is free of this concern by construction.
3. The fork experiment evaluates the gradient at a *calibrated* critic. Under the clip the critic is not calibrated (Theorem 8), so Table 2 describes the instability at the outset of the departure, not the fully developed dynamics. This is precisely why Proposition 13 cannot close the magnitude gap on its own, and why P1 is the decisive experiment.
4. Theorem 8’s divergence is asymptotic. In practice it is bounded by the 500-episode budget and by `max_grad_norm`= 0.5; the claim is about the direction and the absence of a fixed point, not about an attained infinity.
5. Remark 2: the simulated minutes per step differ between the methodology and the current code. Conclusions are robust to the choice, but θ_{fix} in Equation (16) is not.